

Sensor Technology Project (5 ECTS): Accurate positioning from RTK-GNSS and IMU

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Background

GreenGuard develops tractor-mounted autonomous laser weeding systems for row crops. Accurate positioning is the foundation for everything else: a weed found in an image is only useful if we know where it sits to within centimetres. GNSS gives absolute position but at a low rate and with occasional dropouts; an IMU gives orientation and bridges the gaps.

Objective

Combine RTK-GNSS (u-blox ZED-F9P in base/rover configuration) with an IMU through a Kalman filter, producing a continuous and accurate stream of position and orientation that holds up through short GNSS outages.

Hardware

Component	Description
GNSS rover	u-blox ZED-F9P on the moving platform, RTK-corrected from the base
GNSS base	u-blox ZED-F9P as base station, streaming RTCM corrections to the rover
IMU	Inertial measurement unit for roll / pitch / yaw
Compute	Raspberry Pi for logging and filtering
Platform	Moving platform for test runs

Tasks

- Set up the base/rover link and verify RTK performance: time-to-first-fix and fix stability.
- Place the base station on a permanent physical point and record its coordinate once, so that runs from different sessions share the same reference frame.
- Characterise the sensors: GNSS accuracy by solution type, and IMU bias, noise and drift.
- Log GNSS and IMU on a common timebase, using the GNSS PPS signal as the reference clock.
- Implement a Kalman filter (EKF) fusing GNSS position with IMU accelerations and angular rates, estimating IMU biases.
- Handle the lever arm between the GNSS antenna and the IMU, so the filter reports one well-defined point on the machine.
- Evaluate against surveyed reference points and during simulated GNSS outages of 1-10 seconds, horizontally and vertically.

Deliverables

- A working base/rover setup with documented configuration.
- A Kalman filter producing position and orientation at a defined output rate.
- Logged datasets from test runs containing both raw and filtered solutions.
- A defined pose format: timestamp, position in ETRS89 / UTM zone 32N (EPSG:25832), height, roll / pitch / yaw, and uncertainty.
- Code and a short report in a Git repository.

Scope

5 ECTS corresponds to roughly 135 working hours. The filter is demonstrated on a moving platform over a defined test area rather than a full field. A parallel 10 ECTS R&D project uses the same GNSS/IMU set with a camera to build a georeferenced orthomosaic; the pose format defined here is its direct input.